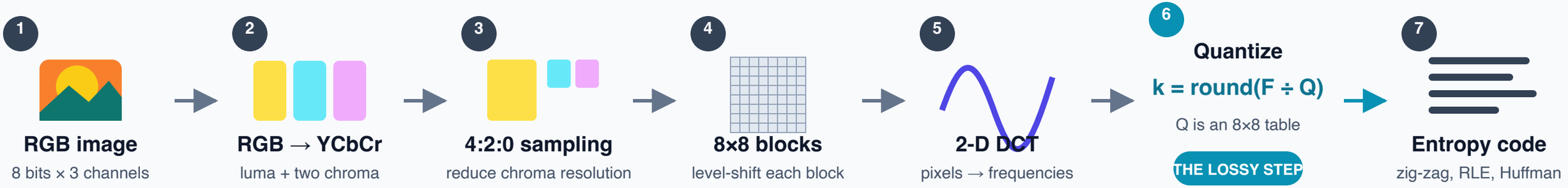
How ordinary JPEG turns pixels into a bitstream

Most stages are reversible engineering choices. Quantization is where information is deliberately discarded.



What the JPEG file contains

headers / metadata	DQT segments the Q tables	Huffman tables	entropy-coded coefficient stream

Decoding reverses the pipeline

Huffman decode → dequantize → IDCT → chroma upsample → RGB

The decoder reads Q directly from the file.

Q — The compression decision lives in 64 numbers per component

Small Q entries preserve a frequency but cost more bits. Large entries discard more detail and usually create more zeros.

Traditional encoders start from fixed, human-designed table shapes and scale them with one quality setting.

X-JPEG changes this box

Q becomes image-adaptive